Operating Systems

CSE Semester 5 — Academic Notes

Synthetic academic knowledge-base document created for CampusAI RAG evaluation and demonstration.

Unit 1 — OS Fundamentals

- An operating system is system software that manages computer hardware and provides services to application programs.
- Major OS responsibilities include process management, memory management, file management, I/O management, protection, security, and resource allocation.
- Kernel mode provides privileged access to hardware, while user mode restricts applications from executing privileged operations directly.
- System calls provide the interface through which user programs request operating-system services.
- OS structures include monolithic kernels, layered systems, microkernels, and modular designs.

Unit 2 — Processes and Threads

- A process is a program in execution with its own address space and execution state. A process control block (PCB) stores information such as process state, program counter, CPU registers, scheduling information, and memory-management information.
- Common process states are new, ready, running, waiting/blocked, and terminated.
- Context switching saves the state of one process or thread and restores the state of another. It enables multitasking but introduces overhead.
- A thread is a lightweight execution unit within a process. Threads of the same process can share code, data, and other resources.
- Inter-process communication (IPC) mechanisms include shared memory and message passing.

Unit 3 — CPU Scheduling

- CPU scheduling selects a ready process to execute on the processor.
- Common algorithms include First-Come First-Served (FCFS), Shortest Job First (SJF), Shortest Remaining Time First (SRTF), Priority Scheduling, and Round Robin.
- Important metrics include waiting time, turnaround time, response time, and throughput.
- Turnaround time = completion time − arrival time. Waiting time = turnaround time − burst time.
- Round Robin assigns each process a time quantum and is commonly used for interactive systems. Very small time quanta increase context-switch overhead.

Unit 4 — Memory Management

- Memory management tracks memory usage and allocates memory to processes.
- Paging divides logical memory into fixed-size pages and physical memory into frames. A page table maps pages to frames.

- Virtual memory allows processes to use an address space larger than physical memory by keeping some pages on secondary storage.
- Page replacement algorithms include FIFO, Optimal, and LRU. A page fault occurs when a referenced page is not currently in physical memory.
- Thrashing occurs when excessive paging leaves the processor with little useful work. Working-set and page-fault-frequency approaches can help control it.

Unit 5 — Deadlocks and File Systems

- A deadlock is a state in which processes wait indefinitely for resources held by one another.
- The four Coffman conditions are mutual exclusion, hold and wait, no preemption, and circular wait.
- Deadlock handling approaches include prevention, avoidance, detection and recovery, and ignoring the problem when appropriate.
- The Banker’s algorithm is a deadlock-avoidance method based on safe states.
- A file system organizes files and directories and manages storage allocation, naming, access control, and metadata.

Quick Revision Checklist

- Know the key definitions and distinguish closely related concepts.
- Practice explaining each concept in your own words before using it in an exam.
- For numerical or algorithmic topics, work through at least one example by hand.
- For database and software-engineering topics, connect definitions to practical examples.

CampusAI RAG metadata suggestion: department=CSE, semester=5, subject=Operating Systems